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PROJECT SYNOPSIS
ON

"G-CARE: An AI Healthcare Assistant"

BACHELOR OF ENGINEERING IN

COMPUTER SCIENCE AND ENGINEERING (AI & ML)

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☐ Accepted ☐ Rejected

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ABSTRACT

G-Care is an AI-based elderly care system designed to improve the safety, independence, and quality of life of senior citizens. With the increasing challenges faced by elderly individuals such as missed medications, lack of continuous health monitoring, and delayed emergency response, there is a need for a simple and effective solution. G-Care addresses these issues by providing smart medicine reminders, basic health monitoring support, and an intuitive user interface tailored for ease of use.

The system enables interaction through a web-based dashboard that allows guardians or caregivers to monitor user activity and respond when necessary. It follows a rule-based methodology in the current phase, where user inputs such as medication schedules are processed to generate timely alerts and notifications. The system ensures that elderly users receive clear reminders and can interact with the application without complexity.

This project is currently developed as a software prototype using modern web technologies such as React and TypeScript. The focus of this phase is on designing the system architecture, implementing the user interface, and establishing the core functionality of reminders and alerts. The prototype demonstrates the feasibility of the proposed solution and lays the foundation for further development.

In future work, the system will be enhanced by integrating wearable devices and sensors for real-time health data collection. Advanced AI and machine learning techniques will be incorporated to analyze health patterns and predict potential risks, enabling proactive healthcare. Additional features such as mobile application support, voice interaction, and cloud-based data management will further improve usability and accessibility.

Overall, G-Care aims to provide a cost-effective, user-friendly, and scalable solution for elderly care, reducing dependency on constant supervision while ensuring safety and timely assistance.

PROBLEM STATEMENT

With the increasing elderly population, ensuring their safety, health management, and independence has become a major concern. Elderly individuals often face difficulties in managing their daily routines, particularly in remembering to take medications on time. Missing or incorrect medication intake can lead to severe health complications and reduced effectiveness of treatment.

In addition, many elderly people live alone or spend long periods without supervision, making it difficult to detect emergencies such as sudden health issues, falls, or critical conditions. The absence of continuous monitoring and delayed communication with caregivers increases the risk during such situations.

Existing healthcare and monitoring systems attempt to address these issues but are often limited by factors such as **high cost, complexity, lack of user-friendly interfaces, and limited accessibility**. Many of these systems require technical knowledge or are not designed specifically for elderly users, making them difficult to adopt. Furthermore, most solutions focus on either monitoring or alerting, without integrating multiple functionalities into a single, simple platform.

There is also a lack of systems that provide **clear visual guidance, intuitive interaction, and direct connectivity with guardians**, which are essential for improving usability among elderly users. Emotional factors such as loneliness and lack of support further highlight the need for a system that not only ensures safety but also enhances overall well-being.

Therefore, there is a need to develop a **cost-effective, easy-to-use, and integrated solution** that can provide timely medication reminders, basic monitoring support, and efficient communication between elderly users and caregivers. The proposed G-Care system aims to overcome these challenges by offering a simple yet effective platform that enhances safety, promotes independence, and ensures timely assistance when required.

OBJECTIVE OF THE PROPOSED PROJECT

The primary objective of the proposed G-Care system is to design and develop a simple, reliable, and user-friendly platform that enhances the safety, health management, and independence of elderly individuals. The system aims to assist users in maintaining proper medication schedules through timely and accurate reminders, thereby reducing the risk of missed or incorrect doses. It also focuses on providing a structured interface that allows guardians or caregivers to monitor user activities and respond effectively when required.

Another important objective is to ensure ease of use by minimizing complexity, making the system accessible even for non-technical users. The project also aims to provide basic alert mechanisms to improve responsiveness during critical situations. Additionally, the system is designed to be cost-effective and scalable so that it can be adopted by a wide range of users.

Furthermore, the project establishes a strong foundation for future enhancements, including integration of wearable devices, real-time health data monitoring, and implementation of artificial intelligence for predictive analysis. These advancements will enable the system to move from basic support to intelligent healthcare assistance, ensuring proactive and efficient elderly care.

- To design a user-friendly system for elderly individuals
- To provide timely medicine reminders to avoid missed doses
- To develop a dashboard for guardians to monitor users
- To ensure basic safety and support through alerts
- To reduce dependency on constant supervision
- To create a cost-effective and accessible solution
- To establish a foundation for future integration of AI and wearable devices

HARDWARE AND SOFTWARE REQUIRED

Hardware Requirements

- Device: Laptop
- Processor: Intel Core i5
- RAM: 8 GB
- Storage: 512 GB SSD
- Internet Connection

Software Requirements

- Operating System: Windows
- Frontend Framework: React.js (Vite)
- Programming Language: TypeScript
- IDE: Visual Studio Code
- Browser: Google Chrome
- Package Manager: npm

EXPECTED OUTCOME

The expected outcome of the G-Care project is the development of a reliable and user-friendly software prototype that addresses key challenges in elderly care. The system will successfully demonstrate the implementation of a smart reminder mechanism that ensures timely medication intake, reducing the risk of missed or incorrect doses. It will also provide a structured dashboard interface that allows easy interaction for users and basic monitoring capability for guardians.

The prototype is expected to validate the feasibility of integrating healthcare support features into a simple web-based application. It will show how alerts and notifications can assist elderly individuals in managing their daily routines more effectively. The system will also highlight the importance of simplicity and accessibility in designing applications for non-technical users.

In addition, the project will establish a clear system architecture and workflow that can be extended in future phases. It is expected to serve as a foundation for advanced functionalities such as real-time health monitoring, wearable sensor integration, and AI-based predictive analysis. These future enhancements will enable proactive healthcare and improved emergency response.

Overall, the G-Care system is expected to contribute towards improving the safety, independence, and quality of life of elderly individuals while providing peace of mind to caregivers through better connectivity and support.

SUMMARY

G-Care is an AI-based elderly care system aimed at improving the safety, independence, and overall well-being of senior citizens. The project focuses on addressing key challenges such as missed medication, lack of continuous monitoring, and delayed response during emergencies. It proposes a simple and user-friendly solution that provides timely medicine reminders, basic alert mechanisms, and a dashboard interface for guardian monitoring.

In the current phase, the project has been developed as a software prototype using modern web technologies. The system demonstrates core functionalities such as reminder generation, user interaction, and basic monitoring flow. The design emphasizes ease of use, making it suitable for elderly individuals with minimal technical knowledge.

The project establishes a strong foundation for future enhancements, including integration of wearable devices, real-time health monitoring, and AI-based prediction for early detection of health risks. Overall, G-Care presents a cost-effective, scalable, and practical approach to improving elderly care through technology, ensuring safer and more independent living.

Student Signature **Guide Signature** **HOD Signature**